

Digital Pollution: Understanding the Energy Costs of Large Language Models

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ABSTRACT

The rapid adoption of large language models (LLMs) has generated growing concern about their environmental costs, including electricity consumption, water use, and carbon emissions. Despite the scale of these impacts, public understanding remains limited, and corporate disclosure practices remain inconsistent. This thesis investigates whether targeted education can improve awareness and shift behavioral intentions toward more sustainable AI use, and whether developer ergonomics, specifically the adoption of small, task-specific models over general-purpose frontier systems, represents the most practical path to reducing AI's energy footprint. The research integrates a literature review, empirical energy benchmarking using direct hardware measurement data from the ML.ENERGY Leaderboard, semi-structured interviews with eight professionals across energy, healthcare, biotech, and machine learning, and two working prototypes: digitalpollution.io, an interactive educational website, and Sentra, a task-specific AI agent for biotech regulatory auditing. The central finding is that large language models are generically powerful but generically expensive. A large reasoning model such as DeepSeek R1 consumes approximately 2.37 joules per token, while a smaller model such as GPT OSS 20B uses only 0.028 joules per token, a difference of nearly two orders of magnitude driven entirely by model selection. Across every interview conducted, professionals independently identified the same conclusion: the answer to AI's energy footprint lies not in rejection, but in specificity. Build small. Build smart. Build with purpose.

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I. INTRODUCTION

Everyone has always been told to think big. But what if, in this case, the answer is to think small?

In an age when our lives are increasingly digitized, the environmental costs of our digital habits often go unseen. Every email, photo, cloud file, and now every interaction with a large language model (LLM) relies on physical infrastructure. Behind every query are data centers filled with servers that consume enormous amounts of electricity and water to train, store, and generate information. Scholars refer to this invisible demand for power as digital pollution [1]. As AI systems grow more capable and more widely adopted, the case for smaller, more efficient, task-specific tools becomes both technically sound and environmentally urgent.

Technology is accelerating discovery across science, education, and industry. As computational tools become more powerful, they also introduce a growing responsibility to use them effectively and thoughtfully. That responsibility extends to everyone: students, corporations, researchers, and developers.

Over the past four years, through research internships at SLAC National Accelerator Laboratory and Stanford University, coursework at Pace University's Pforzheimer Honors College, and conversations with over ten industry professionals across research, energy, biotech, computing, auditing, and compliance, this thesis examines the hidden infrastructure costs behind AI and asks a pointed question: Do people understand the energy and emissions costs associated with large language models? Is there enough accessible information being shared? Can developer ergonomics, specifically the practice of building and training small, task-specific models, shift how AI is used for the benefit of both people and the planet?

This research approaches AI not as a technology to reject, but as one to understand better and use responsibly. The goal is to make the invisible visible: to provide accessible explanations of how LLMs consume energy, and to demonstrate through a working prototype that smaller, scoped models can meet real professional needs at a fraction of the environmental cost.

A. Research Question

The central research question guiding this study is: Do people understand the energy and emissions costs associated with large language models? Are there sufficient publicly available resources on the topic, and can developer ergonomics, such as building and training small models, shift the way AI is used for the benefit of humanity?

B. Scope and Contributions

This thesis integrates three methodological components. First, it compiles and compares published energy benchmarks from open-source models, including BERT, GPT-NeoX, and LLaMA, using direct hardware measurement data from the ML.ENERGY Leaderboard rather than TDP-based estimation. Second, it presents findings from semi-structured interviews with ten professionals spanning energy economics, machine learning,

healthcare AI, biotech regulatory compliance, and developer tooling. Third, it documents the design and development of an interactive educational website and a working AI prototype, Sentra, a task-specific auditing agent for biotech quality and regulatory teams.

II. BACKGROUND

Sustainability has been a personal value of mine for as long as I can remember. I grew up in Arizona and California during the 2010s and experienced firsthand the impact of drought and wildfire. As my family drove between states, the landscape told the story clearly: emptied valleys, scorched hillsides, and in the distance, rows of wind turbines turning slowly against a hazy sky.

Daily conservation practices were part of my upbringing, including reusing water by washing clothes while showering and saving the rinse water from produce for the garden. Being raised in a Buddhist family taught me the importance of interconnectedness, the belief that all humans, animals, and nature are profoundly linked, and that damaging one part of a system harms the whole. These early experiences are my primary motivation for understanding how technological progress can align with ethical and environmental responsibility.

A. Bridging Research and Usability in Renewable Energy Systems

In 2023, I was introduced to a renewable energy initiative within the Grid Integration Systems and Mobility (GISMo) group at SLAC National Accelerator Laboratory. There, I received guidance on translating complex technical systems into accessible resources for researchers and utility developers through website development and Git-based code management. I led publication efforts for multiple energy and sustainability initiatives, creating platforms that shared training materials, simulation tools, and strategic research outputs [12].

Through each project, I became increasingly aware of the hidden complexity behind the scenes. Even as these platforms expanded access to cutting-edge research, the physical systems supporting them. This includes servers, computation, energy use, and cooling infrastructure, which were rarely part of the conversation.

B. From Visualization to Grid Resilience under Extreme Weather

The importance of energy efficiency became more concrete during my work on the Renewable Energy Generation Risk from Outlier Weather (REGROW) project at SLAC National Accelerator Laboratory [10]. REGROW develops reproducible methodologies and data-driven tools to assess vulnerabilities and improve the performance of photovoltaic systems during extreme weather events.

Using Marimo, a reactive Python notebook environment [15], I analyzed meteorological and energy data from the August 2020 California heat wave. I integrated high-resolution data on temperature, solar irradiance, and wind speed to examine anomalous weather patterns and their effects on renewable energy generation. The analyses highlighted the importance of careful system-level optimization, including battery charging and discharging strategies, for maintaining grid stability during periods of high demand.

C. Computational Efficiency and Responsibility in Scientific Discovery

In 2025, my Science Undergraduate Laboratory Internship (SULI) research at the Stanford Synchrotron Radiation Lightsource (SSRL) pushed this perspective further by placing computational efficiency at the center of scientific discovery. I explored methods to automate beamline experiments by integrating real-time analysis with

beamline control systems. Using Python-based automation and a two-detector configuration, the system analyzed fast, low-resolution diffraction data and triggered high-resolution scans only in information-rich regions near diffraction peaks.

Applied to the anatase-to-rutile phase transformation in titanium dioxide (TiO₂), this approach reduced unnecessary background scans and improved time efficiency while preserving data quality for quantitative analysis. This experience marked a turning point in how I think about responsibility in computing: decisions about when to collect data, how much to compute, and which processes to automate carry real environmental implications.

Together, these experiences shape the motivation for this thesis. I approach large language models not as technologies to oppose, nor as tools to adopt uncritically, but as systems whose benefits must be weighed against their growing energy and infrastructure demands.

III. LITERATURE REVIEW

Digital pollution remains underrepresented in broader environmental discussions despite the growing material footprint of digital technologies. Every interaction with a large language model depends on physical infrastructure: data centers, servers, electricity grids, and cooling systems [1]. As AI systems become more deeply embedded in scientific research, business workflows, and everyday productivity tools, their environmental and economic impacts warrant closer examination.

A. Defining Digital Pollution

Digital pollution is best understood as an umbrella concept covering the full lifecycle impact of digital technologies, including e-waste, hardware manufacturing, and the long-term energy demands of data infrastructure [18]. Within this broader category, a more specific and frequently underappreciated phenomenon is internet pollution: the carbon and energy footprint of active data transmission, including streaming video, cloud uploads, and, increasingly, queries to large language models.

This distinction matters for research design because it clarifies which costs are fixed, such as the carbon embedded in manufacturing a GPU, and which scale with user behavior, such as the energy cost of each AI response. Both layers of digital pollution deserve attention, but user behavior has the most direct potential for mitigation through awareness and education.

B. Digital Pollution and Consumer Mitigation Strategies

Common sources of digital pollution include storing undeleted emails, streaming video through cellular networks rather than Wi-Fi, enabling automatic video play on platforms like YouTube, subscribing to unnecessary cloud storage plans, and refreshing social media feeds repeatedly throughout the day [18]. Each of these activities places additional load on server infrastructure, triggering energy consumption that is invisible to the end user but physically real at the data center.

Internet pollution compounds these individual behaviors at scale. A single unread email stored on a server contributes a small but non-zero carbon cost. Multiplied across billions of accounts worldwide, persistent digital clutter becomes a meaningful source of emissions. With approximately 2.3 billion people currently using cloud storage services, the strain on data centers continues to grow [18]. Choosing Wi-Fi over mobile data for internet-intensive tasks can meaningfully reduce the energy cost of data transmission, since mobile networks are generally less energy-efficient per bit transmitted than fixed broadband infrastructure [18].

For consumers seeking to reduce their digital footprint, the literature supports several practical behavioral changes: deleting old emails and attachments, unsubscribing from unused newsletters, disabling autoplay features, favoring Wi-Fi over cellular data, and keeping devices longer to reduce manufacturing-related e-waste [18]. In the context of AI specifically, analogous behaviors include writing concise, targeted prompts; batching related questions into a single query; and selecting smaller, task-appropriate models rather than defaulting to the largest available option [4], [6].

C. Training and Inference in Large Language Models

Large language models are machine learning systems trained on massive text datasets to generate human-like language by predicting word sequences. Two computational processes dominate their energy use: training and inference. Training involves adjusting billions of parameters through repeated computation on large datasets, often requiring hundreds or thousands of GPUs or TPUs to operate continuously for weeks or months. Researchers estimate that training a single state-of-the-art LLM can consume megawatt-hours to gigawatt-hours of electricity, comparable to the annual energy use of hundreds of U.S. households [3], [5].

Inference, by contrast, refers to generating outputs after deployment. While a single inference request requires far less energy than training, inference becomes a major contributor to long-term energy demand when scaled to millions of daily interactions worldwide [6]. Because measurement practices vary and reporting remains inconsistent, non-technical users often misunderstand the distinction between these processes [4], [7].

D. Response Verbosity and the Energy Cost of Every Token

A particularly underappreciated driver of inference energy use is response verbosity. Research in human-computer interaction consistently finds that users prefer longer, more elaborated AI responses, perceiving them as more helpful and authoritative. However, from an energy perspective, every additional word in a model's reply represents one or more additional forward passes through the GPU: each output token requires the model to execute a full computation through its neural network layers [6]. Longer responses, therefore, scale energy consumption nearly linearly with output length.

This dynamic is further amplified by a new class of models known as reasoning or thinking models, such as Qwen 3 Thinking and similar architectures, that generate extended internal chains of thought before producing a final answer. These internal reasoning steps are themselves token-generating processes, meaning that a single user query may trigger tens of thousands of output tokens before the visible response is ever produced. Compared to a standard chat model, reasoning models can consume energy at an order of magnitude greater per request, making model selection a consequential energy decision for users and developers alike.

E. The GPU Bottleneck and the Efficiency Paradox

The energy cost of a single inference token is not fixed. Newer GPU architectures, such as the NVIDIA H100, are considerably more energy-efficient per floating-point operation than older accelerators like the A100. However, this hardware efficiency gain does not automatically translate into lower total energy consumption. Because more capable hardware enables the deployment of more complex and parameter-rich models, the total energy cost per user request frequently increases even as the underlying silicon becomes more efficient.

This phenomenon, known as the Jevons paradox in energy economics, means that measuring AI sustainability requires examining full system-level costs rather than hardware efficiency alone [16]. The ML.ENERGY Leaderboard's direct measurement approach captures this distinction by recording actual power draw during real inference workloads rather than relying on theoretical hardware specifications.

F. Comparing AI to Conventional Search

To contextualize the energy demands of generative AI, it is useful to compare them with conventional web search. A single Google search query consumes roughly the energy equivalent of running a 60-watt lightbulb for 17 seconds. Generative AI responses, which require significantly more computation, can consume substantially more energy depending on the size of the model and the length of the response. Smaller, more efficient models can narrow this gap considerably, while large frontier models widen it.

This comparison offers a relatable reference point for public education. Most users have some intuition that using the internet consumes energy; far fewer understand that replacing a search with an AI-generated response can represent a meaningfully larger energy cost per interaction.



Fig. 1. Energy per output token (joules) across representative open-source models. Source: ML.ENERGY Leaderboard [17].

G. Measuring Energy Consumption: The TDP Myth

Measuring the energy and emissions burden of LLMs remains a significant methodological challenge, frequently distorted by a common misconception known as the TDP Myth. Thermal Design Power (TDP) is the maximum heat a GPU is designed to dissipate, often conflated with maximum power consumption. Many informal energy estimates calculate cost by multiplying a GPU's rated TDP by runtime, assuming the hardware is always operating at its thermal ceiling.

To understand things deeper, let's break down energy per token into its components:

$$\frac{\text{Energy}}{\text{Token}} = \frac{\text{Power} \cdot \text{Time}}{\text{Token}} = \frac{\text{Power}}{\text{Token Throughput}}$$

Real-world benchmarking data from the ML.ENERGY Leaderboard, which directly measures GPU power draw during LLM inference using hardware power sensors, shows that actual power consumption typically ranges from 25 to 75 percent of rated TDP, depending on model architecture, batch size, and workload type [17]. TDP-based estimates can therefore overstate actual energy use by up to a factor of four, with consequences in both directions: alarming the public with inflated figures while simultaneously making efficiency gains look smaller than they are.

The ML.ENERGY Leaderboard addresses this gap by providing direct, hardware-level power measurements for inference tasks across a range of open-source models, reported in joules per output token [17]. For reference, the benchmark shows that a large model such as DeepSeek R1 consumes approximately 2.37 joules per token, whereas a smaller model such as GPT OSS 20B uses only around 0.028 joules per token, a difference of nearly two orders of magnitude that illustrates how dramatically model selection affects energy footprint.

Analyses summarized by MIT Technology Review further explain that credible estimates must specify their system boundaries, such as whether calculations include only accelerator power or encompass full data-center operations, including cooling, redundancy, and power distribution [4]. Because many corporate disclosures remain opaque, ranges and uncertainty intervals are more honest than single-point estimates.

H. Ethical Foundations and Transparency

Training a single LLM can require extensive computational resources, often involving thousands of high-performance accelerators consuming electricity frequently sourced from non-renewable grids [18]. From an environmental perspective, data centers that support AI workloads contribute directly to greenhouse gas emissions, particularly in regions where electricity generation remains dependent on fossil fuels [4]. From a water-use perspective, cooling systems for AI clusters can consume millions to tens of millions of liters of water per year [4]. In water-stressed regions such as California, this level of consumption is comparable to the annual water required to produce thousands of pounds of almonds.

Recent examples illustrate the scale of this expansion. Meta is set to build a Hyperion data center as large as Manhattan as part of an investment of hundreds of billions of dollars. OpenAI is building the Stargate data center in Texas, projected to emit 3.7 million tons of CO2 equivalent per year once operational, roughly equal to the annual emissions of Iceland. xAI has been sued by residents of South Memphis due to air pollution caused by 35 gas turbines powering its Colossus data center. In the Power Hungry Processing presentation, Sasha Luccioni stated, "We have to take back the power, pun intended, from the big AI companies, and put it back into the hands of the developers, regulators, and users of AI" [25]. Luccioni's framing reframes AI sustainability as a power imbalance, not merely a technical problem.

This motivates the small models proposal: by enabling smaller organizations and individual professionals to train and deploy their own specialized models, the field can redistribute both the computational capacity and the economic benefit of AI. Rather than depending on a handful of frontier systems controlled by a few corporations, a diverse ecosystem of purpose-built models returns agency to the people who understand their own domains best.

These concerns align with early ethical critiques of large-scale language models. In *On the Dangers of Stochastic Parrots*, Gebru et al. argue that the dominant research culture rewards scale and performance while sidelining environmental and social costs [2]. Although the paper does not directly measure energy consumption, it emphasizes that limited transparency around training data, compute resources, and system design prevents meaningful accountability.

I. Regulatory Landscape

AI data center environmental legislation is shifting toward stricter oversight of energy and water usage. While federal efforts aim to accelerate permitting for infrastructure, they are increasingly countered by state-level bills focusing on sustainability, transparency, and local grid impacts [26], [27].

At the federal level, the Data Center Community Impact Act, introduced by Rep. Bonnie Watson Coleman (H.R. 7858, 119th Congress) in March 2026, proposes a federal study on the environmental, economic, and public health impacts of data centers on communities [26]. The Artificial Intelligence Environmental Impacts Act of 2024 (S. 3732 / H.R. 7197, 118th Congress), introduced by Sen. Edward Markey, would direct the EPA and NIST to study AI's environmental impacts and establish voluntary reporting standards [28]. The Data Center Transparency Act (H.R. 6984, 119th Congress) would require the EPA to track and report on the water and energy impact of data centers [29].

At the state level, as of early 2026, more than 300 bills had been filed across 30 states focused on data centers, with at least 12 states advancing moratorium legislation to pause new projects while assessing grid, water, and community impacts [27]. States like Maine, New Hampshire, New York, and Oklahoma have introduced moratorium bills, while Illinois suspended data center tax incentives in February 2026 [27].

Internationally, the EU AI Act is considered the world's most comprehensive AI regulation, featuring environmental protection as a core goal with dedicated sustainability rules that also apply to US-based firms offering models in the EU [30]. The Energy Efficiency Directive (EU) 2023/1791 imposes mandatory transparency obligations for data centers with power demand exceeding 500kW, requiring annual reporting on energy consumption, water usage, and renewable energy use [31].

J. Public Understanding and Misconceptions

While technical discussions often focus on infrastructure and algorithms, public understanding of AI's environmental impact remains limited. As documented in *Artificial Intelligence: A Modern Approach* and the *AI Index Report*, publications in machine learning increased by more than twentyfold over the past decade, student

enrollment in AI-related courses rose sixteenfold internationally, and conference attendance at venues such as NeurIPS increased by approximately eight hundred percent [9], [14]. AI has surpassed early expectations, enabling advances in autonomous vehicles, machine translation, speech recognition, medical diagnostics, and climate science.

Alongside these benefits, however, significant environmental risks persist. Recent analyses indicate that approximately seventy percent of new AI-related articles present neutral or positive narratives, with ethical discussions often focusing on data privacy and algorithmic bias rather than environmental impact [14].

Survey data from the Searchlight Institute further illustrate this gap. In some surveys, fewer than half of respondents correctly identify data centers as major contributors to AI's environmental footprint, with individuals from non-technical backgrounds particularly likely to misattribute impacts [7]. At the same time, the survey finds that support for transparency and regulation increases markedly when respondents are informed about AI's energy and water use, suggesting that misinformation stems less from apathy than from limited exposure to clear explanations [7].

Media narratives reinforce widespread misconceptions by framing the cloud as intangible. As MIT Technology Review observes, many people think of the cloud as an invisible thing, when in reality it is very physical: cables, power lines, and energy centers storing and processing everything [4].

K. The Small AI Revolution

A counter movement is emerging. Researcher Sasha Luccioni describes what she calls the small AI revolution: the development of models that are orders of magnitude smaller than traditional frontier LLMs. The smallest are around 135 million parameters, making them approximately 5,000 times smaller than DeepSeek's model. These models are flipping the script by using less data, less compute, less energy, and still having the same level of performance [25].

Luccioni's claim is consistent with findings across multiple interviews in this thesis. Akshay Agrawal described a Stanford team that trained a tiny fake-news detection model on a laptop in minutes and tied for first place against a team using eight GPUs on AWS. The Sentra prototype demonstrates the same principle in a professional context: a lean, retrieval-augmented agent running on low-cost cloud infrastructure performs the specific task of regulatory gap detection without requiring a frontier model's massive compute footprint. The key condition is task specificity. Small models match or exceed large models when the problem is well-scoped, and the training data is curated to the domain. When the problem is open-ended or requires general world knowledge, large models retain an advantage. This thesis argues that most professional AI use cases fall into the former category.

Real-world examples include Rainforest Connection, which uses small AI models for bioacoustic monitoring of illegal logging, running on old cell phones powered by solar energy. Open Climate Fix uses AI to analyze satellite imagery, weather forecasts, and topography data to predict the output of solar and wind installations, supporting decarbonization of energy grids worldwide. Since they are smaller and cheaper to train, they give smaller AI companies the ability to connect with a community and compete with bigger AI companies, because

they can actually afford to train and deploy these models, adapting them to different uses and sharing them back with the community. Proving the reduce, reuse, recycle method also applies to AI [25].

This democratization argument is one of the most important in the thesis. Beyond energy savings, small models offer compounding benefits: data privacy, because sensitive information never leaves the organization's environment; cybersecurity, because a smaller attack surface is easier to audit; and cost, because training and inference on smaller models is accessible to organizations that cannot afford frontier API pricing. Luccioni's reduce, reuse, recycle framing connects AI development to a sustainability ethic that already resonates with environmentally conscious practitioners. The case studies in this thesis, including Nicole Ha's regulatory auditing workflow, Dr. Eevani's provider data management, and Akshay Agrawal's computational caching approach, each demonstrate how a community-oriented, task-specific model can serve a professional need while reducing the broader computational burden.

L. Implications for Research Design

Together, this literature helps pull back the curtain on how AI systems actually work and why their environmental impact deserves attention. Each prompt and response does not emerge from thin air. It travels through warehouses of machines, humming servers, power lines, and cooling systems that quietly draw from finite resources. The gap between what users think they understand about AI's energy use and what they actually know motivates this study's focus on both objective and perceived knowledge. Research demonstrating that awareness can increase support for transparency and regulation informs this thesis's focus on behavioral intentions, such as reducing unnecessary queries, writing more precise prompts, and selecting smaller models for simple tasks.

IV. METHODOLOGY

This thesis integrates three methodological components designed to address the central research question from complementary angles: empirical measurement, qualitative expert interviews, and interactive prototype development.

A. Data Analysis

The data analysis component compiles and compares published energy benchmarks from open-source models such as BERT, GPT-NeoX, and LLaMA to estimate carbon footprints across architectures. This analysis uses direct hardware measurement data from the ML.ENERGY Leaderboard rather than TDP-based estimation, in order to avoid the systematic overestimation that characterizes many informal analyses [17]. The Zeus library is the open-source measurement framework that powers the ML.ENERGY benchmark used to instrument GPU power draw during inference, providing a reproducible and community-validated measurement methodology.

B. Interview Design

Semi-structured interviews were conducted with ten professionals across research, energy, biotech, computing, health insurance, auditing, and compliance. This breadth allows comparison between public understanding and expert perspectives across industries where AI is being actively evaluated or deployed. Participants were recruited through existing professional connections, including industry contacts, guest speakers, and colleagues from prior internships and research positions.

Interviewees include:

- David Chassin, Ph.D., Owner, Eudoxys Sciences LLC; former Chief Scientist, GISMo group, SLAC National Accelerator Laboratory
- Bennet Meyers, Senior Researcher, National Renewable Energy Laboratory (NREL); Adjunct Professor, Electrical Engineering, Stanford University
- Rimvydas Baltaduonis (Rim), Lecturer, Department of Economics, Stanford University
- Akshay Agrawal, Co-founder, Marimo.io; systems engineer and AI researcher
- Arya Boudaie, Software Engineer, Amazon; Adjunct Professor, Pace University; Thesis Advisor
- Jonathan Lee, Clinical Assistant Professor, Computer Science, Pace University
- Dr. Sri Ramesh Eevani, AVP Technology, Healthfirst
- Nicole Ha, Executive Consultant, BioTech regulatory, quality, and CMC compliance

V. INTERVIEW ANALYSIS AND KEY QUOTES

The following section presents synthesized findings from semi-structured interviews conducted with eight professionals between February and April 2026. Each interview was recorded, transcribed, and analyzed for recurring themes related to AI energy use, model selection, developer practices, and policy. Direct quotes are presented with IEEE-style attribution. Analytical commentary follows each quote, grounded in the raw interview notes and connected to the broader literature reviewed in the preceding section.

A. *Akshay Agrawal. Co-founder, Marimo*

Akshay approached energy efficiency from a developer ergonomics angle rather than a purely environmental one, and that framing is what makes his perspective most valuable to this research. He observes that most developers default to large language models because they are easy, not because they are the right tool. He described how a Stanford team trained a tiny, purpose-built model on a laptop in minutes and tied for first place in a fake news detection competition against a team running massive models on eight GPUs on AWS. The lesson found was that smaller, thoughtfully designed models can match large ones for specific tasks, but they require more expertise upfront.

On the product side, Akshay described two sustainability-adjacent features Marimo is developing: computation caching, which skips expensive reruns when code has not changed, reducing strain on both local machines and cloud servers; and WebAssembly notebooks, which run Python entirely in the user's browser without hitting a cloud backend at all. Both were framed as developer ergonomics improvements that happen to reduce energy consumption as a byproduct. He also flagged a structural problem: progress on making models smaller can really only happen in academia, because large companies have every incentive to scale up, not down.

"Large language models have been enormously successful insofar as they have been widely adopted because they are so general. But you can get comparable answers for far less energy cost and monetary cost if you actually just train small classical machine learning models. The hard part is that it requires expertise and time" [33].

Agrawal's observation highlights a core tension in AI adoption: accessibility versus efficiency. Large models lower the barrier to entry; any developer can query an API and get a reasonable result without deep machine learning expertise. But that convenience comes at a cost that is externalized onto the environment and onto the companies paying for compute. The thesis of this research is that organizations with repetitive, well-defined tasks, such as regulatory auditors, healthcare administrators, and energy analysts, are precisely the populations for which small, task-trained models are most appropriate. The data privacy and cybersecurity benefits Agrawal alludes to are compounding advantages: a model trained on internal company documents never exposes that data to a third-party API, reducing both risk and compliance burden. Bias is also reduced when training data is curated from real, domain-specific, human-reviewed sources rather than the broad and noisy corpus of the internet.

"We trained this really tiny model on our laptop in just a few minutes and tied first place. The team we tied with trained some really large models on eight GPUs on AWS. If you are conscious about energy or efficiency, it is good to remember that small models can do really well for specific tasks" [33].

This anecdote is one of the most empirically grounded illustrations in this research. It shows that for a bounded classification task like determining whether two articles agree or disagree, a small, purpose-built model trained on a laptop in minutes can match the performance of a large model trained on eight GPUs over days. The energy difference between these approaches is not incremental; it is orders of magnitude. The Sentra prototype follows this same logic: rather than sending regulatory documents to a frontier LLM for open-ended analysis, it uses a retrieval-augmented pipeline that retrieves only the relevant requirements and runs a structured comparison. The result is faster, more auditable, and dramatically cheaper in both cost and energy than a general-purpose alternative.

"We are thinking about caching computations so that expensive computations do not always have to be rerun. It will benefit users, it will benefit us because there will be less strain on our servers, and it will benefit the environment as a result" [33].

Agrawal's framing of caching as a simultaneous win for users, the product, and the environment is a model for how sustainability can be embedded in tooling without requiring users to make an environmental choice. The energy saving is a byproduct of good engineering, not an ethical demand. This is an important design principle for the educational website: the goal is not to guilt users, but to surface efficient alternatives that are already better on every dimension.

"If you are interested in making models as small as possible, academia is a good place to do that, because they do not have as many resources, they are required to use fewer" [33].

The institutions best positioned to advance small model research are precisely those that lack the compute budgets to compete on frontier model development. This creates a natural alignment of incentives: resource-constrained academic labs are motivated to find efficient architectures, and the field benefits from that work in the form of smaller, more deployable models. This thesis proposes that this research priority needs to be recognized and funded explicitly, not treated as a byproduct of budget limitations.

B. Bennet Meyers. Staff Scientist, SLAC / NREL

Bennet's core argument is that AI is a marketing term more than a technical one, and its application to energy systems is frequently unjustified. He documented five concrete problems: growing energy and water footprints, exploitative labor conditions for data labelers, poor accuracy and hallucinations, inability to audit systems when they fail, and incompatibility with open science. The researcher's most actionable insight is the question every engineer should ask before reaching for AI: Do I actually need this? He argues that convex optimization, control theory, and classical statistics can solve most energy problems more efficiently, more interpretably, and more

sustainably than neural networks. Currently, he applies these methods to time series analysis for solar power forecasting and EV charging optimization using CVXPY, a Python-based convex optimization modeling language.

In our interview, Bennet explained this critique with his de-anthropomorphizing framework, arguing that language like thinks, understands, and decides is not just imprecise but actively harmful. It inflates public expectations, obscures corporate accountability, and makes honest conversations about costs impossible. Beneath the polished interface of every intelligent system is a largely invisible workforce of data labelers, annotators, and system supervisors who are chronically underpaid and uncredited, what Bennet calls brute-force labor hidden by design. The most important social factor he brought up was a growing trend of organizations mandating tools like Microsoft Copilot simply because they have purchased a Microsoft 365 license: usage driven by compliance rather than need, generating unnecessary compute at scale while obscuring both the energy cost and the human cost behind a corporate productivity narrative. He noted this trend extends even to NGOs and non-profit organizations.

When AI is framed as marketing rather than a precise technical category, the implications for energy accountability skyrocket. Vague language allows companies to inflate their AI narratives while obscuring the actual compute involved. It also shapes public perception in ways that make accurate energy estimation harder: a user who believes they are interacting with something that truly thinks is less likely to question whether a simpler tool would serve them better. Bennet's de-anthropomorphizing framework [23] proposes a direct remedy: describe these systems by what users can do with them, not by anthropomorphic capabilities they purportedly possess.

"Replacing well-understood systems with general learning algorithms is a bad goal. Smart, effective use of computation is a good goal" [34].

Meyers is not arguing against computation or even against AI broadly; he is arguing against the reflexive adoption of the most complex available tool for problems that do not require it. His own net load forecasting work using smooth multiperiodic time features, a mathematically elegant, interpretable model that performs well without requiring massive computing, is a direct demonstration. The same principle applies to Sentra: a retrieval-augmented agent running structured comparisons against a curated requirements database is not the most powerful AI system available, but it is the most appropriate one for the task. Smart, effective use of computation means matching the tool to the problem, not defaulting to the largest available option.

"Stochastic automation systems can offer substantial utility but also have the risk of causing serious harm. Hype and imprecise language make this more difficult to navigate. We should describe these systems by what users can do with them, not by anthropomorphic capabilities they purportedly possess, and acknowledge the significant human labor that goes into making these systems work, from data corpus curation and data labeling to system supervision and maintenance" [34].

Meyers's official statement connects directly to Gebru et al.'s Stochastic Parrots argument [2], which identifies the same tension between scale and accountability. Both argue that the AI research community's reward

structures push toward larger, more powerful, more opaque systems while undervaluing the interpretability, auditability, and human labor costs that make systems safe to deploy. Where this thesis extends both arguments is in the practical dimension: it is not enough to critique scale; the field needs existence proofs that smaller, interpretable systems can meet real professional needs. Sentra is one such proof. Meyers's own CVXPY-based energy models are another. The limits of this thesis lie in the fact that Sentra has not yet been benchmarked against a frontier model on the same regulatory auditing task, which would be a valuable extension of this work.

C. Rimvydas Baltaduonis. Energy Economist, Stanford University

Baltaduonis framed AI infrastructure as a multidimensional problem: not just an energy issue, but a security, economic, and policy challenge simultaneously. His most important point is that the race to AI dominance has become a national security concern, meaning governments and private companies are throwing resources at data centers without adequate environmental accountability. Everyone is giving incentives, he noted, but no one has clearly defined who pays for the externalities.

Baltaduonis highlighted that data center development affects communities directly through noise, water use, and local grid strain, and that the people living near these facilities often have no voice in the decision. He brought up concerns about information quality: citizens making decisions about AI policy are increasingly relying on algorithmically curated social media feeds that prioritize engagement over accuracy, making informed democratic participation harder.

"This is a multidimensional problem. It is a security issue. It is a national race to AI competence" [35].

Framing AI energy as a national security issue has a double edge. On one hand, it elevates the stakes and demands government attention beyond what an environmental framing alone might generate. On the other hand, it risks subordinating sustainability concerns to competitiveness concerns: if the dominant frame is that the United States must out-build China in data center capacity, environmental costs may be treated as acceptable collateral. Rim's insight is that this framing is already happening, governments are providing incentives without asking who pays for the environmental consequences, and that economists are better positioned than technologists to model those externalities. This thesis argues that the educational layer is a critical complement: if citizens and policymakers understand the real costs, the political calculus changes.

"It is not just the government. Private companies need land, water, and resources. And the noise issues, the community impacts, everyone is giving incentives, but no one is asking who pays" [35].

The infrastructure of AI disproportionately burdens communities that did not choose to host it and do not capture the economic benefits [26]. The xAI lawsuit by South Memphis residents, cited in the Ethical Foundations section, is a real-world example. The economic externality is not abstract; it is measured in air quality, water availability, noise levels, and grid strain in specific neighborhoods. Rim's

point is that the accounting framework has not caught up to the infrastructure expansion, and that without transparency requirements and community consent mechanisms, the burden will continue to land on those with the least political power.

"Citizens are not very informed. Social media pre-feeds narratives, conspiracy profiles, clickbait, and information not from trusted sources. That is who is making decisions about AI regulation" [35].

The power of education and awareness comes hand in hand. "Citizens are not very informed. Social media pre-feeds narratives, conspiracy profiles, clickbait, and information not from trusted sources. That is who is making decisions about AI regulation" [35]. This is the most direct validation of the educational website component of this thesis. If the information environment that shapes public opinion on AI regulation is dominated by engagement-optimized content rather than accurate technical or economic reporting, then a well-designed, accessible, interactive educational tool represents a meaningful intervention.

If the information environment that shapes public opinion on AI regulation is dominated by engagement-optimized content rather than accurate technical or economic reporting, then a well-designed, accessible, interactive educational tool represents a meaningful intervention. The website's design, anchoring abstract statistics in relatable comparisons, using real benchmark data rather than inflated TDP estimates, and giving users the ability to simulate their own AI workloads, is specifically intended to be the kind of trustworthy, clear resource that Rim identifies as missing from the current information landscape.

"Micro-grids, solar plus battery plus data center, could be economically sensible. But we are using diesel generators to substitute when there is not enough renewable power. Pollution is increasing for communities near these centers" [35].

The Carnegie Mellon techno-economic analysis Baltaduonis references suggests that islanded micro-grids pairing solar, battery storage, and data centers are already economically viable; the barrier is not technology or economics but policy. This is a critical finding: the solutions exist, but the regulatory framework that would incentivize them is being weakened at the same moment that AI infrastructure demand is accelerating. The thesis cannot resolve this political tension, but it can argue that transparency, education, and the adoption of small models are levers that operate independently of federal policy and that remain available regardless of EPA enforcement status.

D. Arya Boudaie. Software Engineer, Amazon; AI Professor, Pace University

Arya Boudaie provided both technical depth and a regulatory overview that grounds this thesis in current policy developments. As a software engineer at Amazon and the thesis advisor for this project, his perspective helped bridge industry practice and academic inquiry.

Boudaie noted that LLMs exhibit jagged intelligence, excelling at complex tasks while failing simple ones, and that general-purpose frontier models are not the right tool for every application [36]. Tool use inside LLMs remains rudimentary, and the field has significant room for efficient, task-specific design. An example from his work: a content moderation task that required checking whether a body of text was under 200 words was being handled by an LLM, which was counting tokens one by one using a full GPU inference pass. Running an optical character recognition model or a simple character-counting script would have accomplished the same task at a fraction of the energy cost.

Boudaie's Amazon example is particularly instructive because it comes from inside a major technology company deploying AI at scale. An LLM is not better at counting words than a two-line Python script; it is just the tool that was most available and easiest to reach for. Jagged intelligence means that LLMs create an illusion of general capability that can lead practitioners to apply them to tasks for which they are poorly suited and computationally wasteful. The correction is not better prompting but better task decomposition: identify what the system actually needs to do, and match the tool accordingly.

E. Jonathan Lee. Clinical Assistant Professor, Computer Science, Pace University

Professor Jonathan Lee offered a behavioral economics framing of AI energy use that complements the technical and policy perspectives gathered from other interviewees. His central observation is that most users do not think about the energy costs of AI tools for the same reason they do not think about the labor conditions behind their clothing: the cost is not paid at the point of consumption.

"If you are not paying the sticker price, or someone else pays, you do not really think about the energy costs. In the same way that people buy clothing with questionable provenance without thinking about labor conditions or materials sourcing" [37].

Lee's analogy connects AI energy consumption directly to Pigou's externality framework [16]: when the true social cost of a product is not reflected in its price, consumers will overconsume it. The AI energy market is a textbook example. Frontier models are priced on a subscription or per-token basis that does not include the cost of cooling water, grid strain, or community air quality impacts near data centers. Without price signals that reflect these costs, users have no economic incentive to choose more efficient tools. This is precisely why Lee is skeptical that education alone can shift behavior at scale. His written response suggests that organized campaigns or price reform are more likely to produce systemic change than awareness alone. This thesis takes the position that education is a necessary precondition for effective advocacy and regulation, even if it is not sufficient on its own

Lee also noted that most consumers default to the best free model available, and that providers tend to offer leaner models for free while keeping the highest-power models behind a paywall. This creates a market structure that, paradoxically, may already be nudging some users toward lower-energy models without their awareness. He also raised the case of Anthropic's most capable models being too costly to release broadly, suggesting that energy and infrastructure constraints are already shaping product decisions in ways the public does not see.

F. David Chassin, Ph.D. Owner, Eudoxys Sciences LLC; Former Chief Scientist, GISMo, SLAC

David Chassin offered one of the most methodologically precise perspectives in this thesis. As the owner of a consulting firm serving utilities, regulators, and government agencies, and a former chief scientist at SLAC, he brings decades of experience applying machine learning to electric power systems. Notably, he declines to use the term AI altogether, preferring to describe his methods in terms of the specific techniques used: time-series general additive models, white-box statistical approaches, and on-premise computing. He reasons that AI refers to computational problems not yet solved, and that once a problem is solved, it should be described by the method that solved it.

Chassin uses on-premise and desktop computing systems for most of his work because the data he processes is often highly sensitive and cannot be loaded onto cloud platforms. This has an accidental sustainability benefit: desktop-based inference is extremely energy-efficient compared to cloud GPU inference, and the white-box statistical models he uses are orders of magnitude leaner than LLMs. He tracks training wall-clock time as a proxy for energy consumption and prioritizes reducing it, though he does not track energy use directly.

"The biggest problem I see with AI is whether it is being used appropriately to solve various problems and whether the applications justify the social, political, and economic costs. There is very little evidence to show that they are useful for anything beyond predicting the next word in a sentence. The only job that seems to be at risk is the copy editor. Everything else is too important to risk handing off to a machine that is best characterized as a stochastic parrot" [38].

Chassin's position is the most skeptical in this thesis, and it is worth taking seriously. His invocation of the stochastic parrot framing [2] is not a rhetorical flourish; it reflects a practitioner's frustration with the gap between AI claims and demonstrated utility in high-stakes domains. His work on electric grid resilience, EV charging optimization, and energy forecasting demonstrates that well-understood mathematical tools, convex optimization, general additive models, and probabilistic methods can solve these problems with interpretable, auditable results and without the computational overhead of large neural networks. The thesis does not fully adopt Chassin's skepticism; Senra is, after all, an AI agent, but it does adopt his core principle: match the method to the problem, choose interpretability when stakes are high, and do not mistake marketing for technical capability.

Chassin's condition on reporting is a sophisticated policy argument. Energy data without social and economic context is easy to greenwash: a company can report low energy per query while obscuring the volume of queries, the communities displaced by its data center siting decisions, or the labor conditions of its annotation workforce. His point is that aggregate data center consumption - already partially captured by utility reporting requirements - may be more useful for accountability than firm-level voluntary disclosure, unless that disclosure is comprehensive. This connects to Rim's observation that the regulatory framework for AI energy has not kept

pace with infrastructure expansion, and to Arya's summary of the legislative landscape in which transparency bills remain proposed rather than enacted.

"I would like to see more rigorous oversight of data center operations, a better model for allocating the infrastructure costs of building data centers so that we do not socialize those costs unnecessarily, and a stronger corporate governance model and legal framework for managing the social, political, and economic risks to society" [38].

Chassin's call to stop socializing infrastructure costs aligns with Rim's externality framing and with the broader argument of this thesis that the current AI economy does not price its true costs. His three-part recommendation: rigorous oversight, accurate cost allocation, and stronger governance - maps to the three intervention levels that this thesis addresses: data (the ML.ENERGY Leaderboard and TDP Myth correction), behavior (the educational website and developer ergonomics proposal), and structure (the Sentra prototype as a model for task-appropriate AI design).

G. Dr. Sri Ramesh Eevani - AVP Technology, Healthfirst [24]

Ramesh Eevani's work at Healthfirst represents one of the clearest examples in this thesis of AI being used in a targeted, human-centered way. His team focused on two specific problems: reducing documentation burden in clinical settings and transforming how provider data is validated and updated.

For clinical documentation, he developed an ambient AI scribing system trained on clinician personas, allowing doctors to spend more time with patients rather than taking notes. For provider data management, he transformed manual review of National Provider Identifiers and registrations using intelligent automation, flagging missing NPIs before claim rejections and detecting duplicates automatically.

During the COVID-19 pandemic, this approach was tested under pressure. Healthfirst needed to rapidly onboard temporary providers, bypassing certain credentials to expedite the process. The turnaround time for onboarding dropped from 90 days to 3 weeks across 13 primary sources of verification. The New York Times recognized Healthfirst for this achievement. His design philosophy is orchestration first and automation second: map the workflow, then automate specific steps, rather than replacing the entire process at once.

Eevani's orchestration-first philosophy is a direct counterexample to the big-bang AI deployment approach that Bennet Meyers critiques. Rather than replacing clinical documentation or provider management wholesale with a general AI system, Eevani identified specific, bounded subtasks - scribing ambient speech, flagging missing NPI numbers, detecting duplicates - and applied targeted automation to each. This is precisely the developer ergonomics argument applied in practice: the energy cost of a system that does one thing well is dramatically lower than the energy cost of a frontier model processing every possible variation of a healthcare workflow. The 90-to-21-day onboarding reduction was achieved not by deploying more powerful AI but by deploying more precisely scoped AI.

H. Nicole Ha - Executive Consultant, BioTech Regulatory, Quality, and CMC

Nicole Ha provided the professional context that motivated the Sentra prototype. As an executive consultant helping biotech companies prepare for FDA and EMA inspections, her day-to-day work involves reviewing and writing controlled documents, understanding what companies actually do, writing procedures, and representing clients before regulators to defend their processes.

The problem she identified is straightforward: the manual work of cross-referencing regulatory requirements against company documentation is slow, error-prone, and expensive. An AI agent that understands regulatory law, interprets how it applies to specific company processes, and surfaces evidence gaps before an audit would transform this workflow entirely. This is the use case Sentra was designed to address.

Nicole's use case illustrates both the value of a task-specific model and the data privacy argument that makes one essential. Regulatory documents, including SOPs, batch records, CAPAs, validation reports, and IND filings, contain highly sensitive proprietary information that companies cannot and should not upload to a third-party general-purpose API. A small model trained on anonymized regulatory frameworks and company-specific document structures, running on internal infrastructure, solves the privacy problem while also being dramatically more accurate for the specific task than a frontier model with no domain training. The security benefit is not incidental: it is one of the primary reasons a specialized tool is more appropriate here than a general one.

VI. DEVELOPMENT AND PROTOTYPE

A. Prototype Development

The prototype development component encompasses two deliverables. First, an interactive educational website built with HTML, CSS, and JavaScript visually simulates how a model processes a prompt, where computation occurs, and how energy accumulates with repeated use. Second, Sentra, a task-specific AI agent for biotech regulatory auditing, serves as a working demonstration of the small models proposal. Both are described in detail in the Development and Prototype section of this thesis.

B. Educational Website

The proposed website is designed as an interactive platform that helps users understand the infrastructure, energy requirements, and financial costs behind AI systems. Rather than only explaining digital pollution, the site allows companies, developers, and researchers to simulate how different AI workloads translate into real-world computing resources.

Users can input parameters such as the type of AI model, the number of users interacting with the system, the number of prompts processed, and the type of hardware used. The platform then estimates the energy required to generate those outputs, how long the system would take to process them, the number of machines needed to support the workload, and the approximate operational costs.

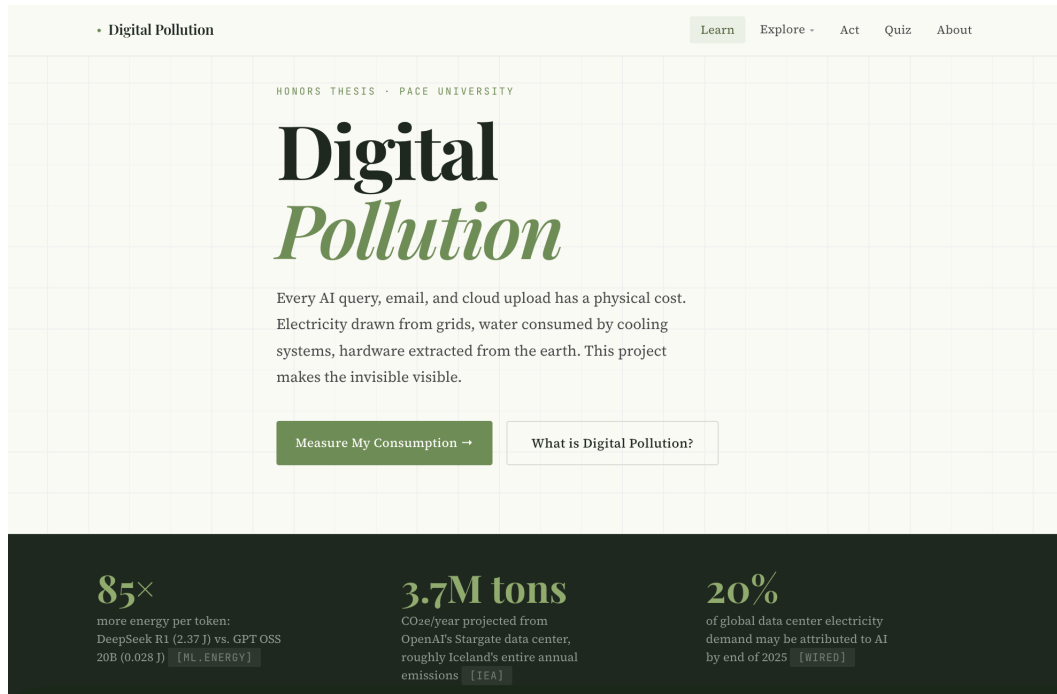


Fig. 2. Educational website homepage. Available at digitalpollution.io

Three core interactive features are planned. The Model Comparison Slider draws on ML. ENERGY Leaderboard data allows users to select and compare different models side by side to see the energy cost per response. For example, a large reasoning model such as DeepSeek R1 consumes approximately 2.37 joules per token, while a smaller model such as GPT OSS 20B uses only around 0.028 joules per token [17]. Visualizing this contrast on a sliding scale makes the otherwise abstract concept of model size tangible and personally relevant.

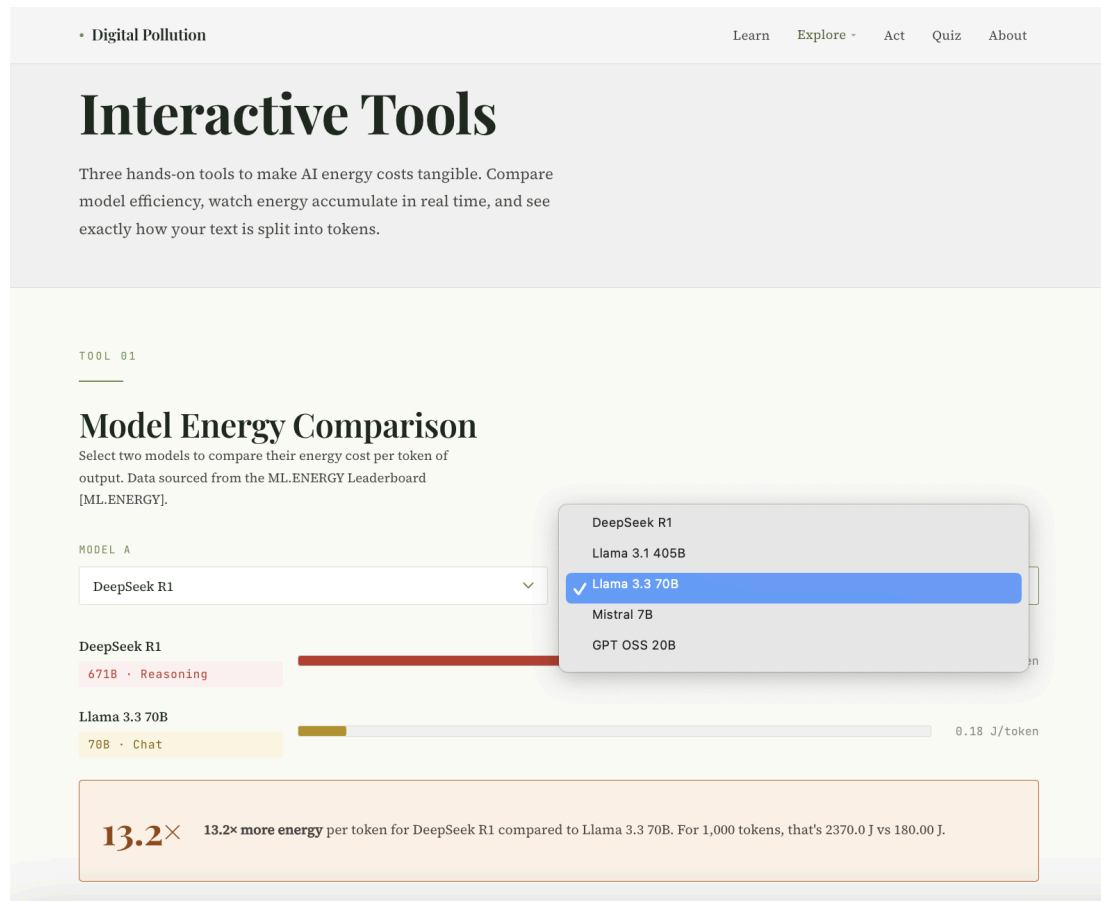


Fig. 3. Model Comparison. Users toggle between model sizes to see real-time energy cost per token from ML.ENERGY Leaderboard data [17].

The Email vs. AI Calculator lets users explore the energy equivalences between familiar digital habits and AI use. If deleting 1,000 stored emails saves a measurable amount of server energy, how many AI prompts does that correspond to? Grounding AI energy costs in everyday digital pollution vocabulary bridges the gap between abstract data and personal behavior.

The Live Energy Meter Widget uses the Zeus library in a Python-based cloud environment to display the estimated energy cost of calling a model in real time. As users type a prompt, the widget animates the accumulating energy cost token by token, making the forward-pass nature of inference visually intuitive. This widget is designed specifically to address the TDP Myth by displaying real measured values rather than theoretical maximums.

• Digital Pollution

LearnExplore ▾ActQuizAbout

TOOL 02

Live Energy Meter

Type a prompt and watch the energy cost accumulate token by token, just as it would on a real GPU. Uses measured J/token values from the ML.ENERGY Leaderboard [ML.ENERGY].

MODEL

Llama 3.3 70B — 0.18 J/token ▾

YOUR PROMPT

The environmental cost of AI is often invisible.

Estimated output: ~28 tokens

Simulate Response

Reset

28

tokens generated

5.040

joules consumed

1.40e-3

watt-hours

✓ Complete — Llama 3.3 70B used 5.040 J for this response.

Fig. 4. Live Energy Meter Widget. Energy cost accumulates in real time as the user types, using Zeus-measured joule-per-token values [17].

Sentra: A Task-Specific AI Agent for Biotech Regulatory Auditing

Sentra is an AI-native operating layer for biotech quality, regulatory, and CMC teams that sits on top of existing document systems. The product continuously maps requirements to evidence across SOPs, batch records, validation reports, and submissions, then identifies gaps, inconsistencies, and risks before audits or filings. It was built because current systems store documents but do not reason across them, forcing teams to manually prepare for audits and submissions, which is slow, error-prone, and expensive. Sentra turns that into a real-time readiness view with clear actions, making regulatory execution measurable rather than reactive.

Technically, Sentra ingests documents into structured storage using Postgres and S3, builds a requirement-evidence graph, and runs task-specific AI agents, including gap detection, conflict detection, change impact analysis, and controlled drafting using retrieval-augmented generation rather than open-ended chat. A

lightweight workflow engine tracks actions and approvals with audit trails, while a React and Next.js interface surfaces readiness scores, risks, and reports. The stack is intentionally lean: Python and FastAPI backend, vector search via pgvector, and API-based LLMs, designed to ship quickly, validate with real customers, and scale into a defensible platform.

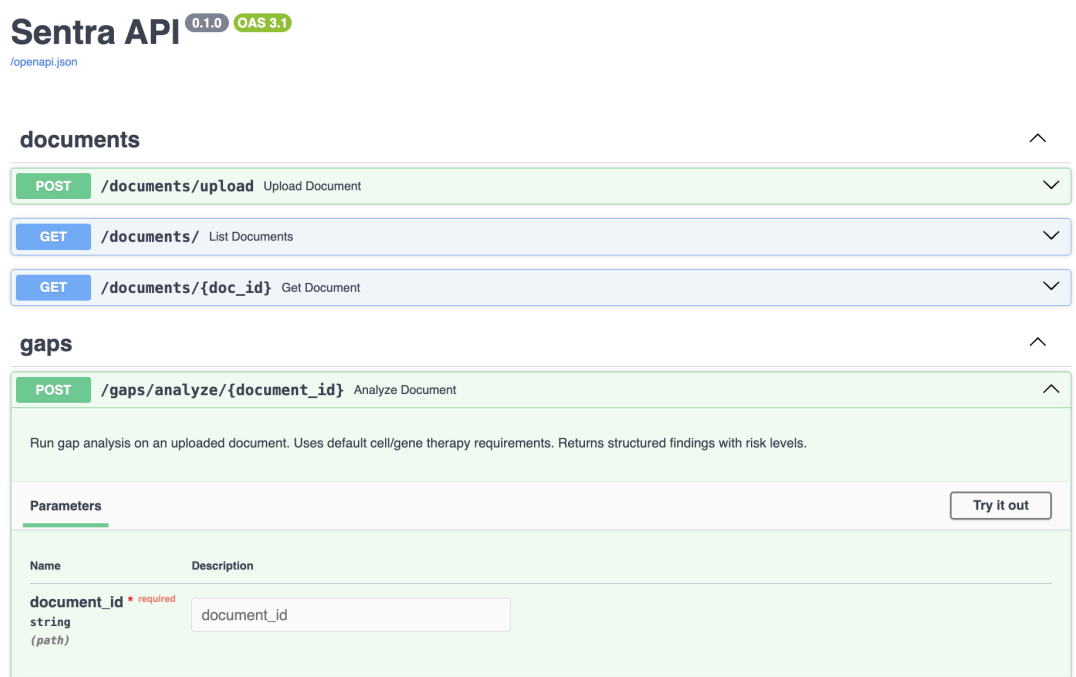


Fig. 5. Sentra API documentation. Live at sentra-ebp0.onrender.com/docs. GitHub: [melodyxnguyen/sentra](https://github.com/melodyxnguyen/sentra).

The system is live at sentra-ebp0.onrender.com/docs and available on GitHub at [melodyxnguyen/sentra](https://github.com/melodyxnguyen/sentra). The database is hosted on Supabase with six tables: documents, requirements, evidence links, findings, actions, and AI outputs. Every AI call logs model version, prompt version, inputs, and outputs, forming an audit trail consistent with GxP compliance requirements. Every output is flagged as requiring human review, positioning Sentra as a decision-support tool rather than an autonomous decision-maker.

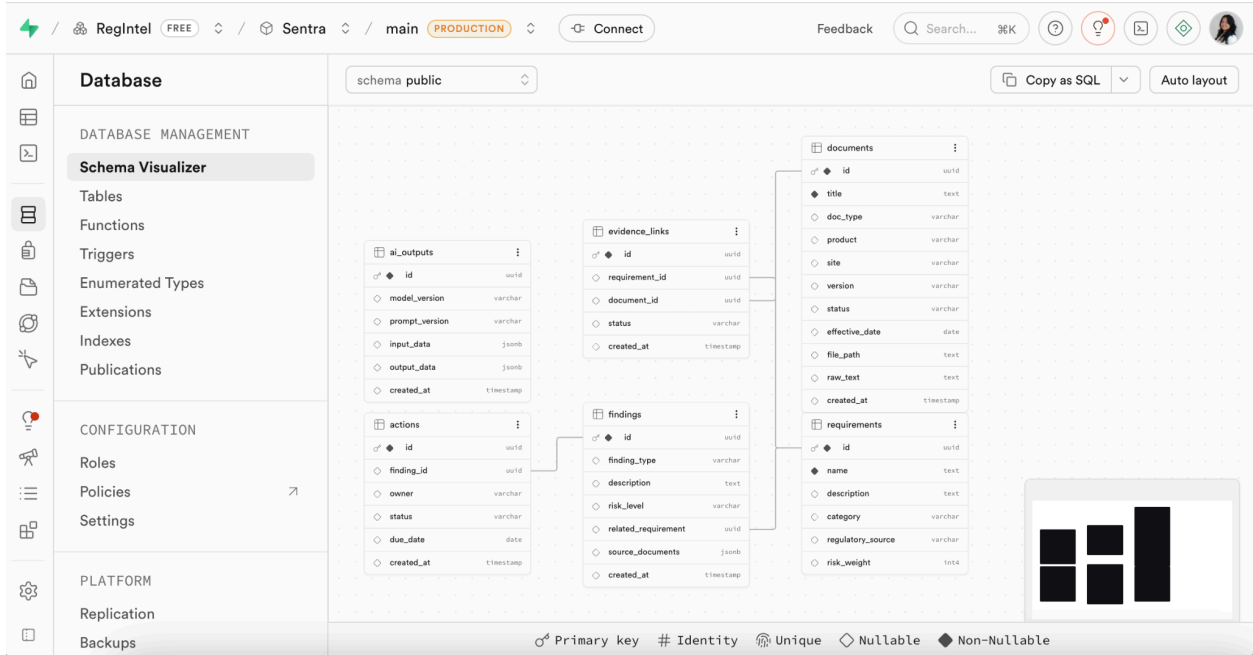


Fig. 6. Sentra database schema hosted on Supabase. Six tables support document ingestion, requirement mapping, gap detection, and audit trail logging.

Sentra demonstrates the core thesis argument in two ways. First, it is a working existence proof that a task-specific AI agent can be built on a student budget, deployed to a live production environment, and perform meaningful regulatory analysis. Second, its architecture reflects the developer ergonomics principles advocated throughout this thesis: retrieval-augmented generation rather than open-ended chat, structured JSON outputs rather than free text, human review required for every output, and a lean stack that avoids unnecessary compute. The primary limitation of this prototype is the absence of a direct energy comparison with a frontier model performing the same task. Future work should use the Zeus library to measure the actual GPU power draw of Sentra's inference pipeline alongside a frontier alternative such as GPT-4o, providing the empirical grounding that this thesis's benchmarking section provides for general model comparisons.

VII. BROADER IMPACTS

The environmental costs of digital technologies disproportionately affect low-income and developing regions, which often host the data infrastructure of wealthier nations. Presberger and Bernauer find that wealthy countries tend to shift pollution-intensive industrial processes to lower-income nations, creating patterns of ecological inequality in which economic dependence and environmental degradation reinforce each other [26].

Raising awareness about LLM energy consumption contributes to ongoing discussions on AI ethics and sustainability. As Gebru et al. emphasize, unchecked model scaling threatens both social equity and environmental stability [2]. By connecting ethical AI practices to measurable energy metrics, this project situates digital pollution within the movement for responsible innovation and climate justice. Its outcomes are intended to encourage developers and consumers to make informed, sustainable choices in their digital practices.

WIRED reporting suggests that AI may account for up to 20 percent of global data-center electricity demand, with projections indicating substantial increases by the end of 2025 [8]. Although experts caution that precise attribution remains difficult due to limited corporate disclosure, the direction of growth is clear, and the stakes are material.

CONCLUSION

The central finding of this thesis is that large language models are generically powerful but generically expensive. The answer to their energy footprint lies not in rejection, but in specificity. People who perform the same tasks repeatedly, whether as biotech auditors, healthcare administrators, data scientists, or energy researchers, should consider building small, task-specific AI models tailored to their exact needs. These models are more efficient in time and energy, more accurate for specialized tasks, and offer stronger data privacy and cybersecurity benefits by keeping sensitive information local.

This argument is supported by literature showing an 85-fold difference in energy consumption between a large reasoning model and a smaller open-source model performing comparable tasks, by interviews with eight professionals who independently identified the same mismatch between general AI tools and specific professional needs, and by the Sentra prototype, which demonstrates that a lean, task-specific agent can be built on a student budget and deployed in a real production environment.

The educational website completes the picture by translating these findings into accessible, interactive tools that help everyday users understand what their AI interactions cost and what they can do differently.

The future of AI is not bigger models. It is smarter ones.

As a personal reflection, what surprised me most throughout this research is how much my thesis question evolved. It began with a broad interest in pollution from gas emissions and data storage, narrowed to the energy costs of large language models specifically, and ultimately arrived at a conclusion I did not expect: the answer is not rejection, but intentionality. Build small. Build smart. Build with purpose.

Upon graduating, I will be joining Healthfirst's Provider Data Management Team as a Business Analyst and Product Owner, where I hope to apply the foundations explored in this thesis to real-world healthcare challenges. Healthfirst is a not-for-profit health insurer serving nearly two million members across New York City and surrounding communities, with a growing investment in AI-driven healthcare solutions. In this role, I will work at the intersection of technology and operations, translating complex business needs into data-driven systems that improve how provider information is managed, verified, and delivered. As generative AI continues to reshape the healthcare industry, I am particularly interested in exploring how large language models can be deployed responsibly within clinical and administrative workflows, with careful attention to the energy costs, governance, and ethical considerations that this thesis has examined. This opportunity represents not just a career milestone, but a continuation of the questions I have been asking throughout my undergraduate research: What does it mean to build AI systems that are not only powerful, but sustainable and equitable for the communities they serve?

The interviews I conducted with industry professionals gave me the most original and grounded insights of the entire project. Every conversation shifted something. I noticed a pattern across each of them: regardless of

industry or role, each professional independently emphasized the importance of building niche, task-specific tools rather than relying on generalized models. That convergence became the backbone of my argument.

As AI simultaneously reshapes every industry, I will be continuing my education through Georgia Institute of Technology's Master of Science in Computer Science program, with hopes of diving deeper into these studies. To give back to the community, I will also serve as a Lead Instructor with iD Tech at Stanford, teaching Artificial Intelligence, Machine Learning, Python, and AI Agents to the next generation of builders. I hope to frame those classes around responsible, human-centered practice, emphasizing an understanding of limits and concise, intentional usage.

What this research ultimately taught me is that the most important question is not how powerful we can make these tools, but how responsibly we choose to use them. I am grateful to be entering a field where that question still feels open, and where the answer still depends on the people willing to ask it.

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Most of all, thank you to Arya Boudaie. You introduced me to tools I did not know I needed, pushed me to think bigger about my own potential, and encouraged me to apply to Georgia Tech. I will carry this work with me into the Master's program this fall.

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